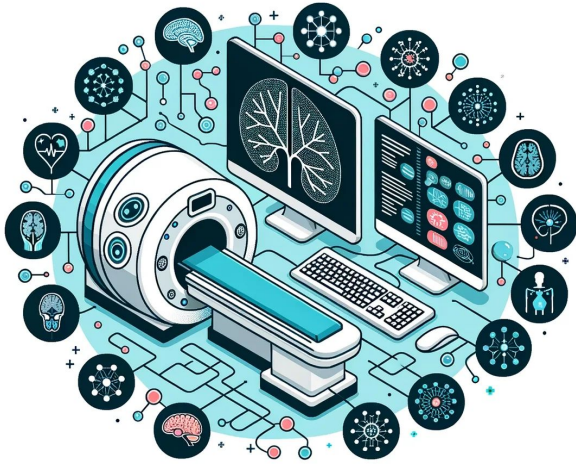


# Applied Deep Learning and Generative Models in Healthcare



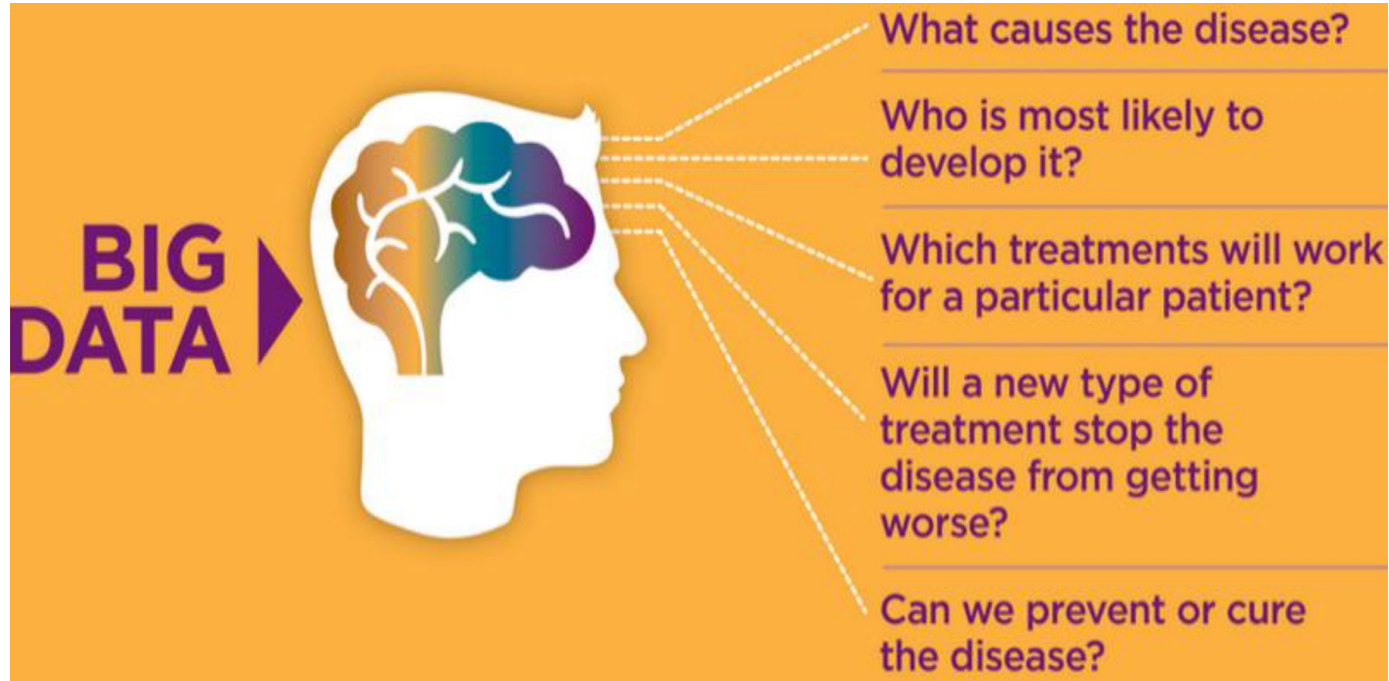
**Session 10:** Automated diagnosis using deep learning

**Date:** Mar 22 2025

**Instructor:** Mahmoud E. Khani, Ph.D.

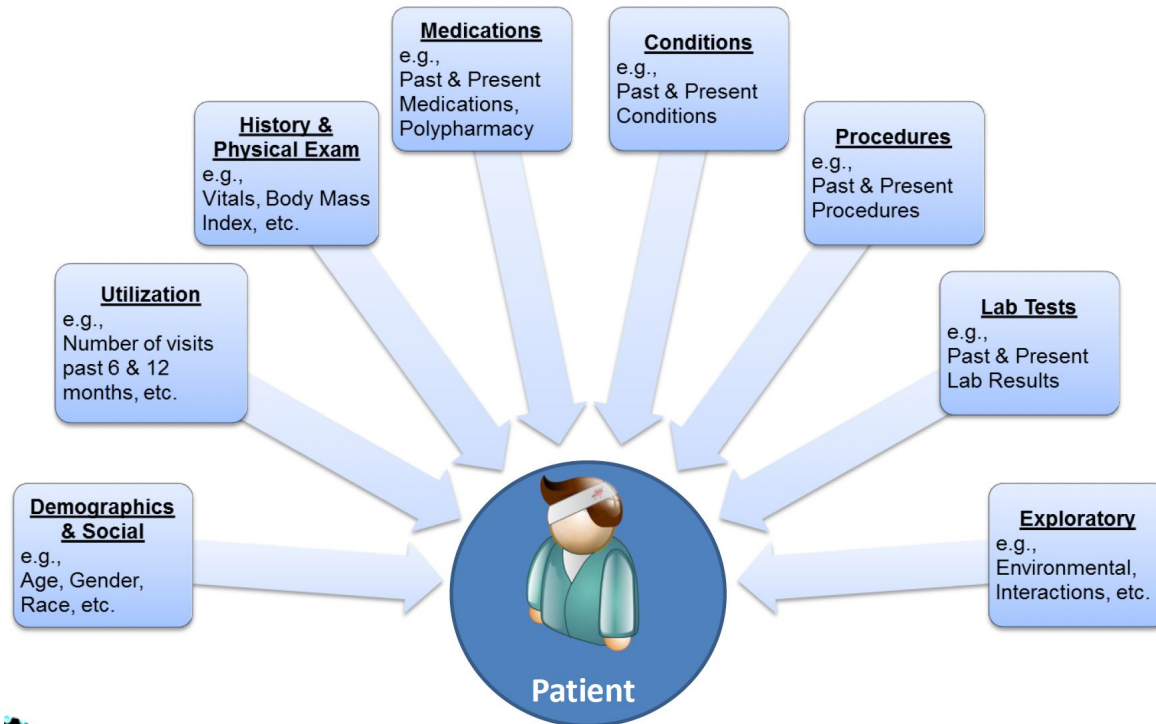
# Big data in healthcare

- There are so many **fundamental questions** to answer!



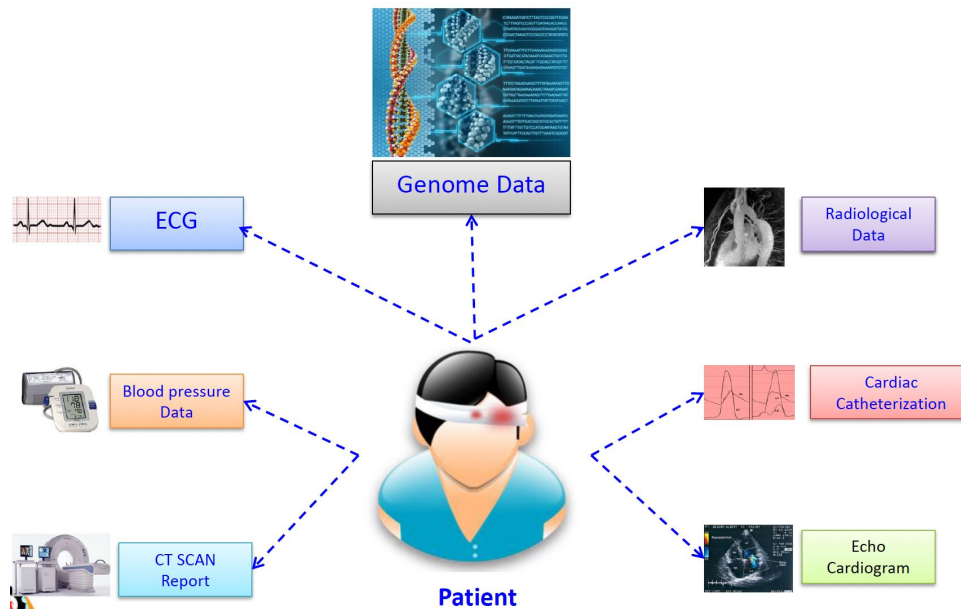
# Structured medical data

- Data collection **does not require imaging**

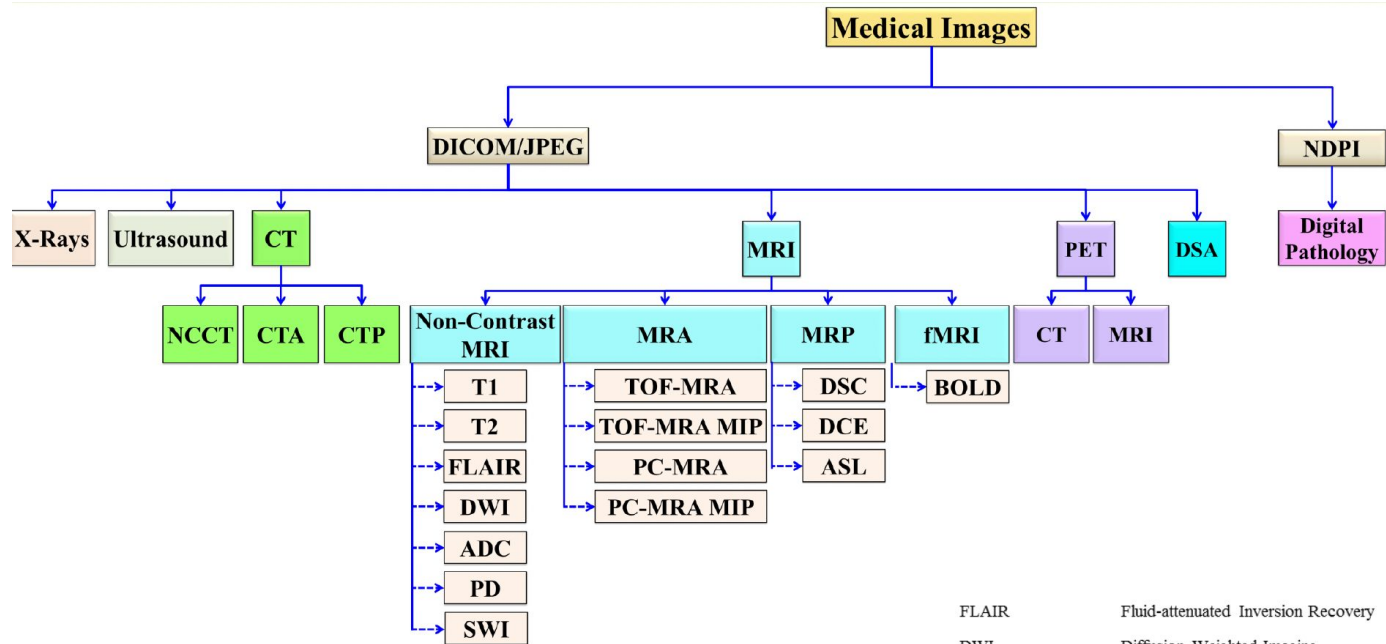


# Unstructured multimodal clinical data

- ECG, CT, MRI, genomic sequences, etc.
  - typically grouped under **medical imaging and signals** or **unstructured clinical data**.



# Types of medical images



CT Computed Tomography  
 NCCT Non-contrast CT  
 CTA CT Angiography  
 CTP CT Perfusion  
 MRI Magnetic Resonance Imaging

MRP Magnetic Resonance Perfusion  
 DSC Dynamic Susceptibility Contrast  
 DCE Dynamic Contrast Enhanced  
 ASL Arterial Spin Labelling  
 BOLD Blood-oxygen-level Dependent  
 PET Positron Emission Tomography  
 DSA Digital Subtraction Angiography

FLAIR Fluid-attenuated Inversion Recovery  
 DWI Diffusion Weighted Imaging  
 ADC Apparent Diffusion Coefficient  
 PD Proton Density  
 SWI Susceptibility Weighted Imaging  
 MRA Magnetic Resonance Angiography  
 TOF-MRA Time-of-Flight MRA  
 TOF-MRA MIP TOF-MRA Maximum Intensity  
 PC-MRA Phase-contrast MRA

# Deep learning is here to help!

- Using sound to diagnose **tuberculosis**



# Deep learning is here to help!

LETTERS

<https://doi.org/10.1038/s41591-019-0447-x>

nature  
medicine

Corrected: Author Correction

## End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography

Diego Ardila <sup>1,5</sup>, Atilla P. Kiraly<sup>1,5</sup>, Sujeeth Bharadwaj<sup>1,5</sup>, Bokyung Choi<sup>1,5</sup>, Joshua J. Reicher<sup>2</sup>, Lily Peng<sup>1</sup>, Daniel Tse <sup>1\*</sup>, Mozziyar Etemadi <sup>3</sup>, Wenxing Ye<sup>1</sup>, Greg Corrado<sup>1</sup>, David P. Naidich<sup>4</sup> and Shravya Shetty<sup>1</sup>



# Deep learning is here to help!

## ARTICLES

<https://doi.org/10.1038/s41551-019-0487-z>

nature  
biomedical engineering

There are amendments to this paper

## Detection of anaemia from retinal fundus images via deep learning

Akinori Mitani<sup>1\*</sup>, Abigail Huang<sup>1</sup>, Subhashini Venugopalan<sup>2</sup>, Greg S. Corrado<sup>1</sup>, Lily Peng<sup>1</sup>, Dale R. Webster<sup>1</sup>, Naama Hammel<sup>1</sup>, Yun Liu<sup>1,3</sup> and Avinash V. Varadarajan<sup>1,3</sup>

Owing to the invasiveness of diagnostic tests for anaemia and the costs associated with screening for it, the condition is often undetected. Here, we show that anaemia can be detected via machine-learning algorithms trained using retinal fundus images, study participant metadata (including race or ethnicity, age, sex and blood pressure) or the combination of both data types (images and study participant metadata). In a validation dataset of 11,388 study participants from the UK Biobank, the meta-data-only, fundus-image-only and combined models predicted haemoglobin concentration (in  $\text{g dl}^{-1}$ ) with mean absolute error values of 0.73 (95% confidence interval: 0.72–0.74), 0.67 (0.66–0.68) and 0.63 (0.62–0.64), respectively, and with areas under the receiver operating characteristic curve (AUC) values of 0.74 (0.71–0.76), 0.87 (0.85–0.89) and 0.88 (0.86–0.89), respectively. For 539 study participants with self-reported diabetes, the combined model predicted haemoglobin concentration with a mean absolute error of 0.73 (0.68–0.78) and anaemia an AUC of 0.89 (0.85–0.93). Automated anaemia screening on the basis of fundus images could particularly aid patients with diabetes undergoing regular retinal imaging and for whom anaemia can increase morbidity and mortality risks.



# Preparing data for automated diagnosis

- Importance of **standardizing datasets** (e.g., LDCT scans, ECG, EEG, MRI, Genomic Data).
- **Data preprocessing techniques**
  - **Normalization & Augmentation**: Improving model robustness.
  - **Segmentation Models (U-Net, nnU-Net)**: For accurate ROI identification.
  - **Data Cleaning & Annotation**: Handling missing data, noise, and bias.

# Building and training deep learning models

- **Common architectures**

- **CNNs**: for image-based diagnosis
- **RNNs**: for sequential data like ECG/EEG
- **Transformer models**: Handling complex data patterns (genomics)

- **Feature extraction techniques**

- **3D convolutions**: For volumetric analysis (e.g., MRI, CT)
- **Hybrid models**: Combining multiple architectures

- **Training best practices**

- **Transfer learning**: Using pre-trained models
- **Cross-validation and hyperparameter tuning**: Enhancing model performance

# Evaluation and model validation

- **Key metrics for diagnosis models**

- **Accuracy, Precision, Recall, F1-Score**
  - Classification performance
- **AUROC (Area under the receiver operating characteristic curve)**
  - General performance
- **Dice coefficient, IoU (intersection over union):**
  - Segmentation quality

- **Model validation**

- **Cross-validation:** Ensuring robustness
- **External validation:** Using independent datasets.

# Confusion matrix

|            |          | PREDICTED LABEL |                |
|------------|----------|-----------------|----------------|
|            |          | NEGATIVE        | POSITIVE       |
| TRUE LABEL | NEGATIVE | TRUE NEGATIVE   | FALSE POSITIVE |
|            | POSITIVE | FALSE NEGATIVE  | TRUE POSITIVE  |

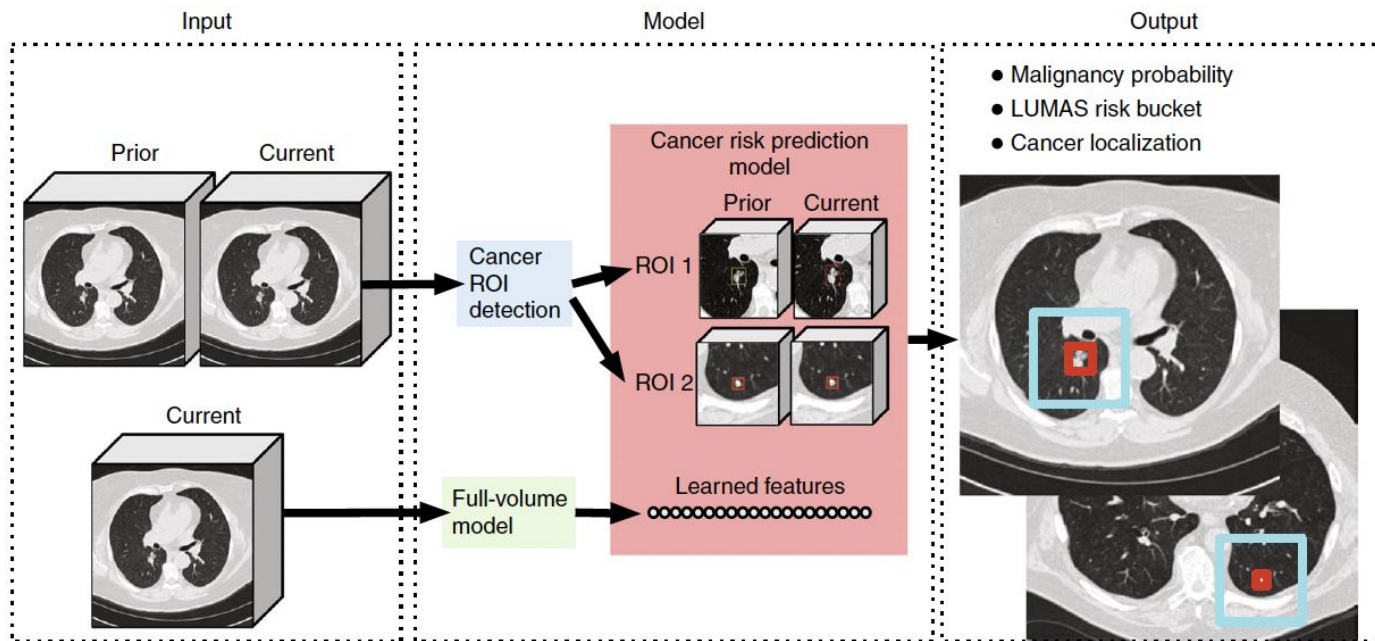
|            |          | PREDICTED LABEL                                                                                       |                                                                                                       |
|------------|----------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|            |          | NEGATIVE                                                                                              | POSITIVE                                                                                              |
| TRUE LABEL | NEGATIVE | <br>TRUE NEGATIVE  | <br>FALSE POSITIVE |
|            | POSITIVE | <br>FALSE NEGATIVE | <br>TRUE POSITIVE  |

# Classification evaluation metrics

Sources: [4][5][6][7][8][9][10][11] view · talk · edit

|                  |                               | Predicted condition                                                                                                                         |                                                                                                 |                                                                                                                                                                                                |                                                                                                                              |
|------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                  |                               | Predicted positive                                                                                                                          | Predicted negative                                                                              | Informedness, bookmaker informedness (BM)<br>$= \text{TPR} + \text{TNR} - 1$                                                                                                                   | Prevalence threshold (PT)<br>$= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$            |
| Actual condition | Total population<br>$= P + N$ |                                                                                                                                             |                                                                                                 |                                                                                                                                                                                                |                                                                                                                              |
|                  | Positive (P)<br>[a]           | True positive (TP),<br>hit <sup>[b]</sup>                                                                                                   | False negative (FN),<br>miss,<br>underestimation                                                | True positive rate (TPR),<br>recall, sensitivity (SEN),<br>probability of detection, hit rate,<br>power<br>$= \frac{\text{TP}}{P} = 1 - \text{FNR}$                                            | False negative rate (FNR),<br>miss rate<br>type II error <sup>[c]</sup><br>$= \frac{\text{FN}}{P} = 1 - \text{TPR}$          |
|                  | Negative (N) <sup>[d]</sup>   | False positive (FP),<br>false alarm, overestimation                                                                                         | True negative (TN),<br>correct rejection <sup>[e]</sup>                                         | False positive rate (FPR),<br>probability of false alarm, fall-out<br>type I error <sup>[f]</sup><br>$= \frac{\text{FP}}{N} = 1 - \text{TNR}$                                                  | True negative rate (TNR),<br>specificity (SPC), selectivity<br>$= \frac{\text{TN}}{N} = 1 - \text{FPR}$                      |
|                  |                               | Positive predictive value (PPV),<br>precision<br>$= \frac{\text{TP}}{\text{TP} + \text{FP}} = 1 - \text{FDR}$                               | False omission rate (FOR)<br>$= \frac{\text{FN}}{\text{TN} + \text{FN}} = 1 - \text{NPV}$       | Positive likelihood ratio (LR+)<br>$= \frac{\text{TPR}}{\text{FPR}}$                                                                                                                           | Negative likelihood ratio (LR-)<br>$= \frac{\text{FNR}}{\text{TNR}}$                                                         |
|                  |                               | False discovery rate (FDR)<br>$= \frac{\text{FP}}{\text{TP} + \text{FP}} = 1 - \text{PPV}$                                                  | Negative predictive value (NPV)<br>$= \frac{\text{TN}}{\text{TN} + \text{FN}} = 1 - \text{FOR}$ | Markedness (MK), deltaP ( $\Delta p$ )<br>$= \text{PPV} + \text{NPV} - 1$                                                                                                                      | Diagnostic odds ratio (DOR)<br>$= \frac{\text{LR}^+}{\text{LR}^-}$                                                           |
|                  |                               | $F_1$ score<br>$= \frac{2 \text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$ | Fowlkes–Mallows index (FM)<br>$= \sqrt{\text{PPV} \times \text{TPR}}$                           | Matthews correlation coefficient (MCC)<br>$= \frac{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$ | Threat score (TS),<br>critical success index (CSI), Jaccard index<br>$= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$ |

# End-to-end lung cancer screening



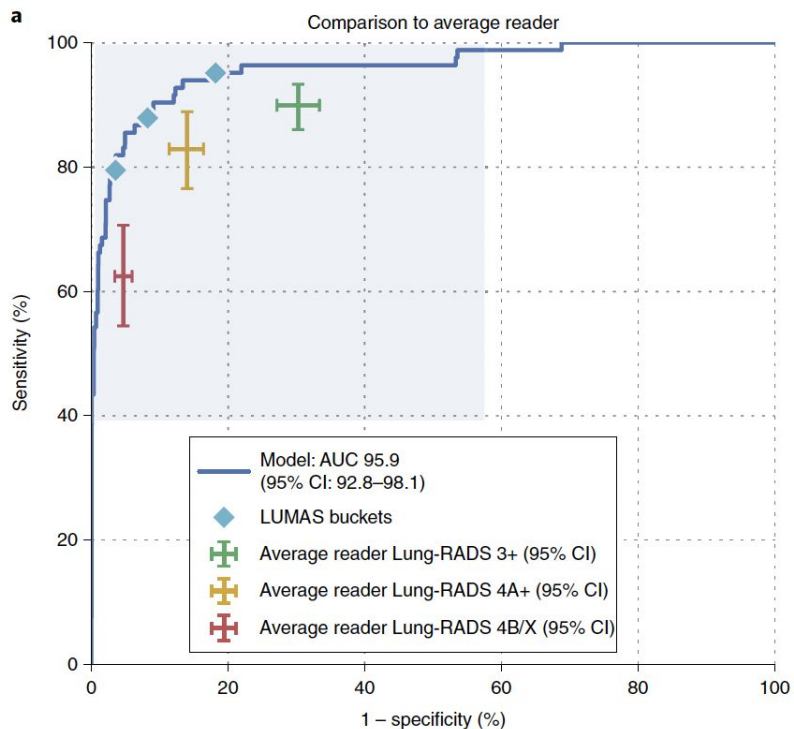
**Fig. 1 | Overall modeling framework.** For each patient, the model uses a primary LDCT volume and, if available, a prior LDCT volume as input. The model then analyzes suspicious and volumetric ROIs as well as the whole-LDCT volume and outputs an overall malignancy prediction for the case, a risk bucket score (LUMAS) and localization for predicted cancerous nodules.



# End-to-end lung cancer screening

- multi-step process, involving segmentation, detection, feature extraction, and prediction.
  - Preprocessing
    - Converting CT scans to standardized input format.
  - Segmentation & ROI Detection
    - Using models to locate suspicious regions (nodules).
  - Feature Extraction & Diagnosis
    - Feeding ROIs to a cancer risk prediction model.
  - Output
    - Generating malignancy probabilities, LUMAS risk scores, and nodule localization.

# Lung cancer screening results



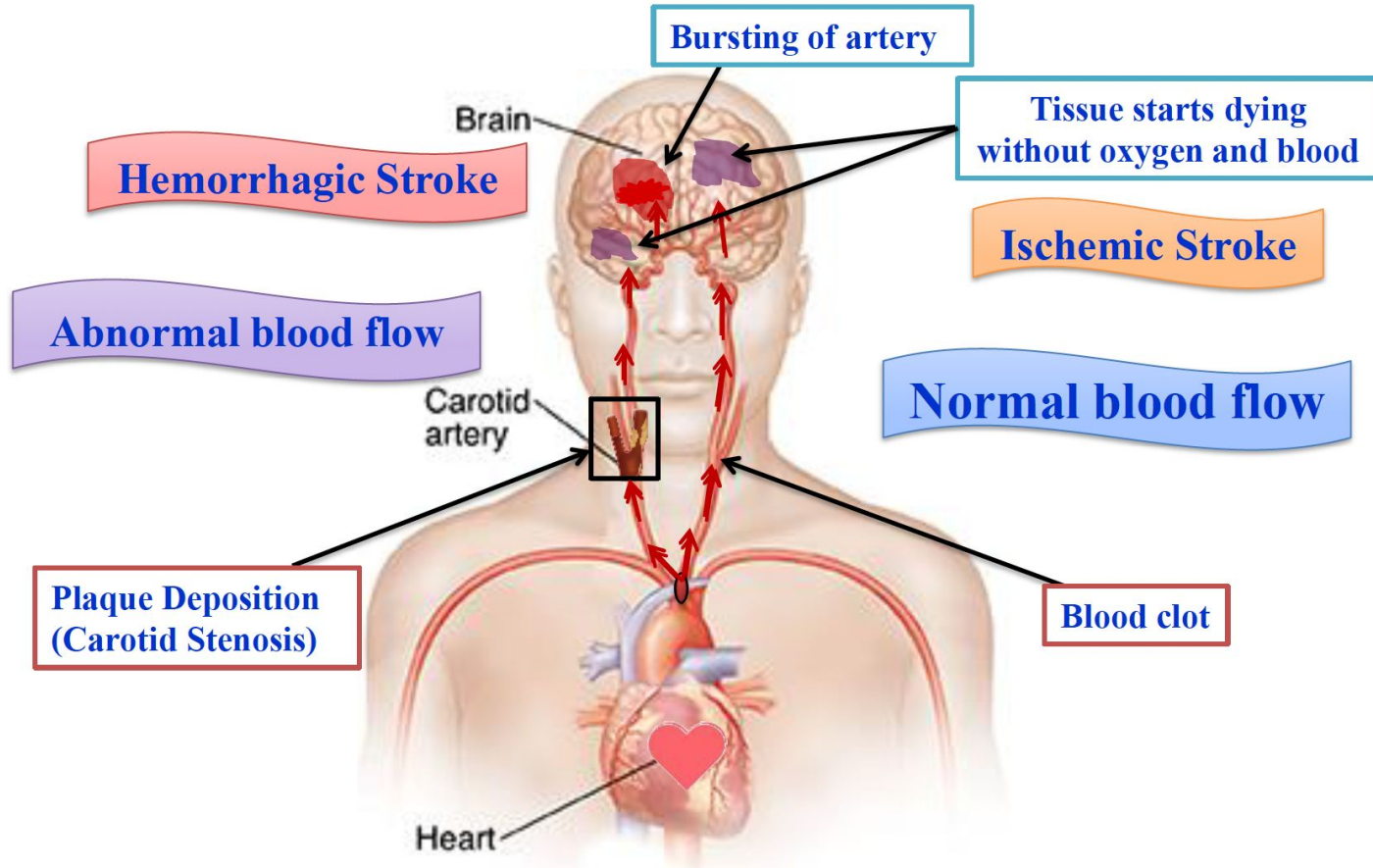
**c**

| Risk buckets               |                | Sensitivity          | Delta                                 |
|----------------------------|----------------|----------------------|---------------------------------------|
| 1,2<br>versus<br>3+        | Average reader | 90.0<br>(86.1, 93.4) | +5.2*                                 |
|                            | Model          | 95.2<br>(89.9, 98.9) | (.4, 9.8)<br>$P = 0.0386$             |
| 1,2,3<br>versus<br>4A+     | Average reader | 82.9<br>(76.6, 89.4) | +7.4*                                 |
|                            | Model          | 90.4<br>(83.3, 96.3) | (1.7, 12.9)<br>$P = 0.0114$           |
| 1,2,3,4A<br>versus<br>4B/X | Average reader | 62.5<br>(54.4, 70.7) | +17.1*                                |
|                            | Model          | 79.5<br>(70.8, 88.2) | (9.0, 24.5)<br>$P < 1 \times 10^{-4}$ |

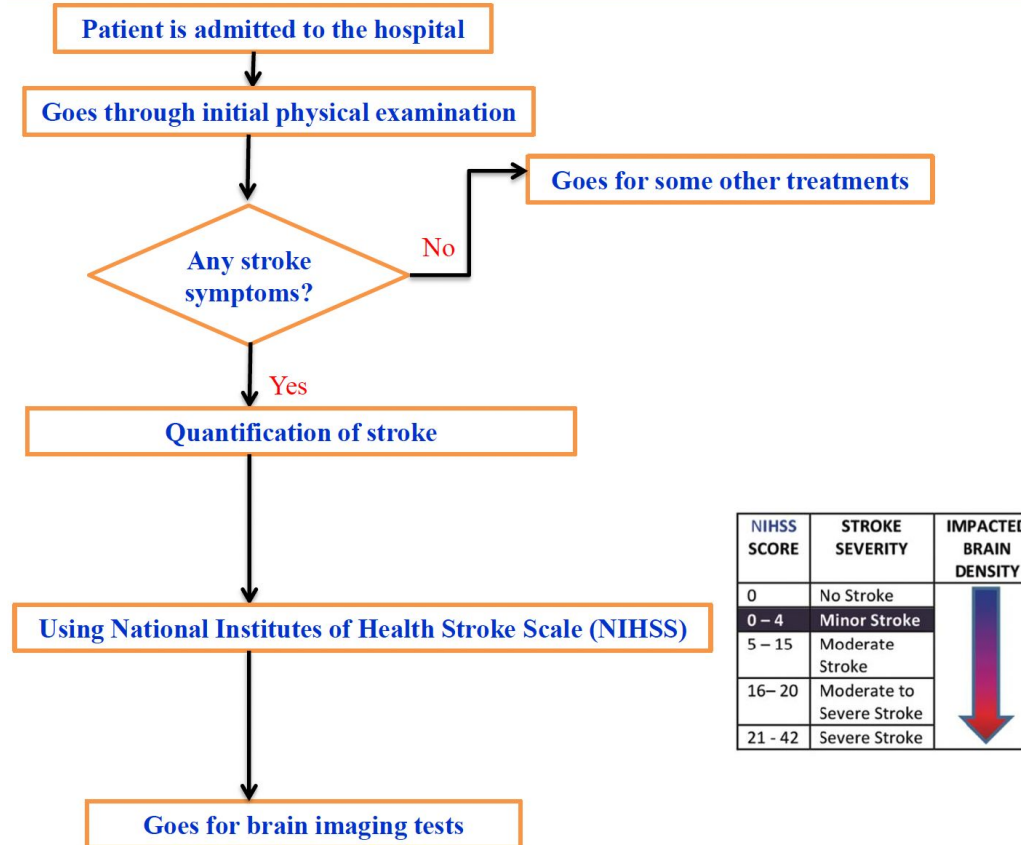
**d**

| Risk buckets               |                | Specificity          | Delta                                 |
|----------------------------|----------------|----------------------|---------------------------------------|
| 1,2<br>versus<br>3+        | Average reader | 69.7<br>(66.6, 72.8) | +11.6*                                |
|                            | Model          | 81.3<br>(77.3, 84.9) | (7.8, 15.1)<br>$P < 1 \times 10^{-4}$ |
| 1,2,3<br>versus<br>4A+     | Average reader | 86.0<br>(83.4, 88.4) | +5.0*                                 |
|                            | Model          | 91.0<br>(88.1, 93.9) | (1.7, 12.9)<br>$P = 0.0008$           |
| 1,2,3,4A<br>versus<br>4B/X | Average reader | 95.3<br>(94.0, 96.6) | +1.1*                                 |
|                            | Model          | 96.5<br>(94.6, 98.2) | (-0.4, 2.6)<br>$P = 0.143$            |

# Cerebrovascular disease

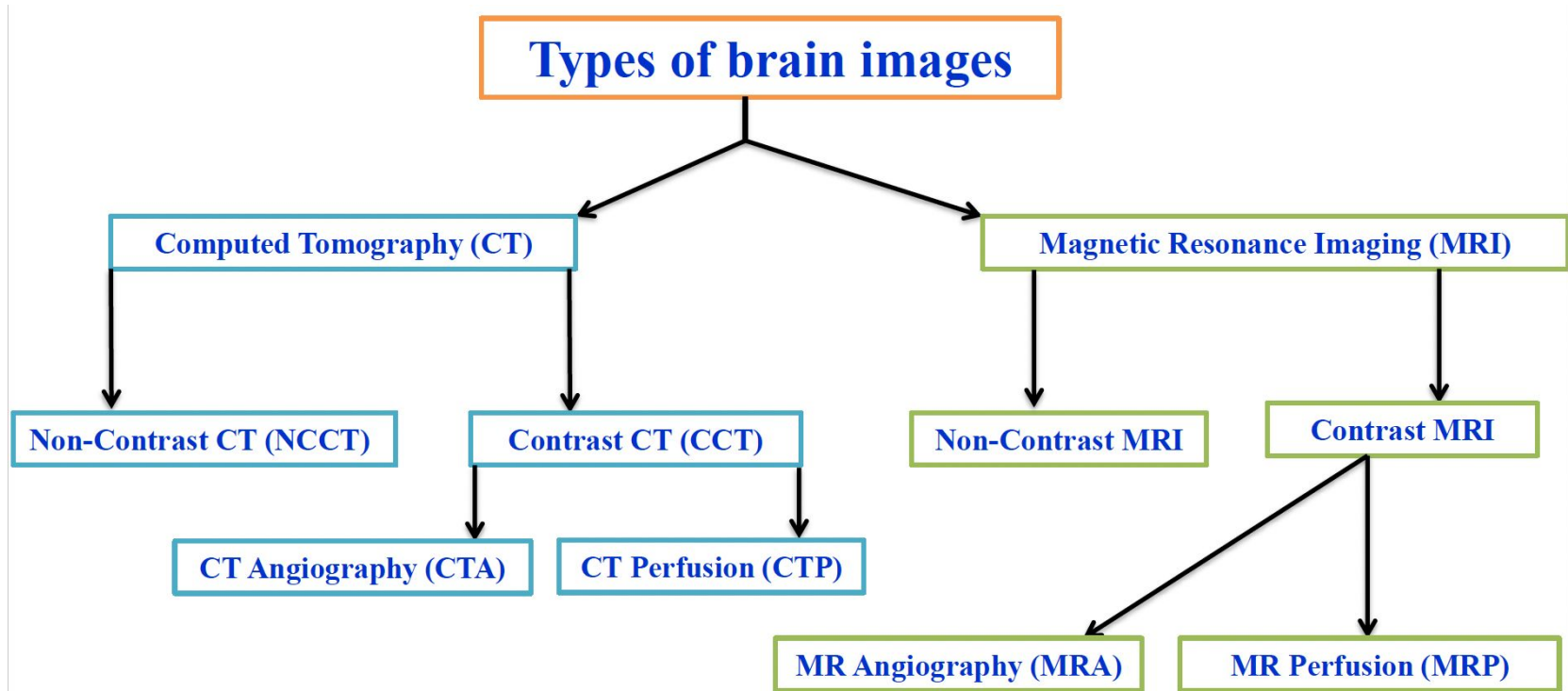


# Diagnosis methods



# Types of brain images

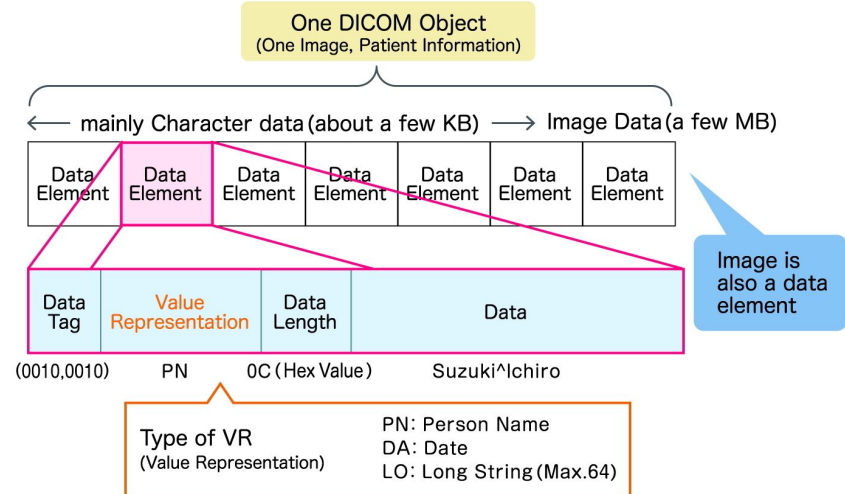
- External **iodine material** used for contrast



# Common brain image format

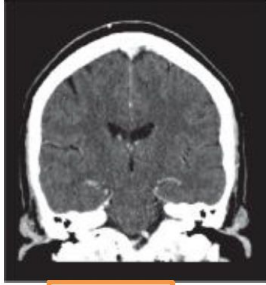
CT/MRI

DICOM (Digital Imaging and Communications in Medicine)

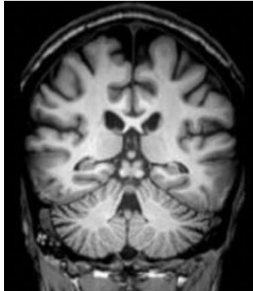


# Imaging planes

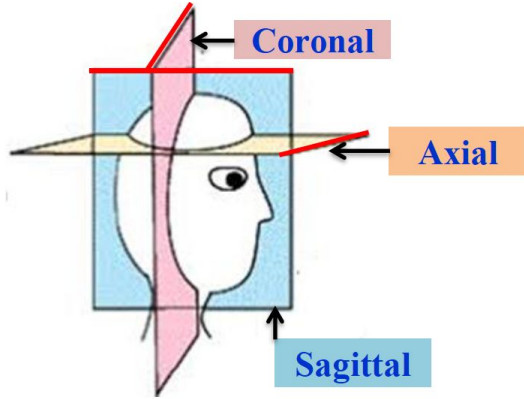
**Coronal**  
(front and back view)



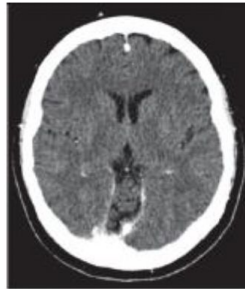
NCCT



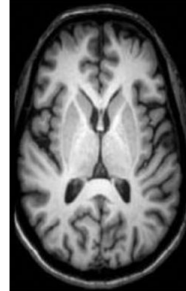
MRI



**Axial (horizontal view)**

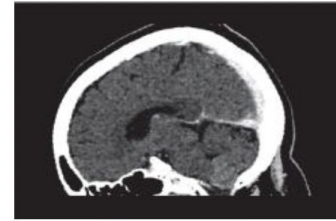


NCCT



MRI

**Sagittal (lateral view)**



NCCT

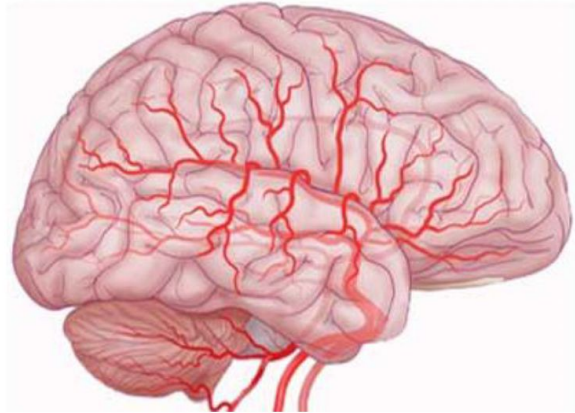


MRI

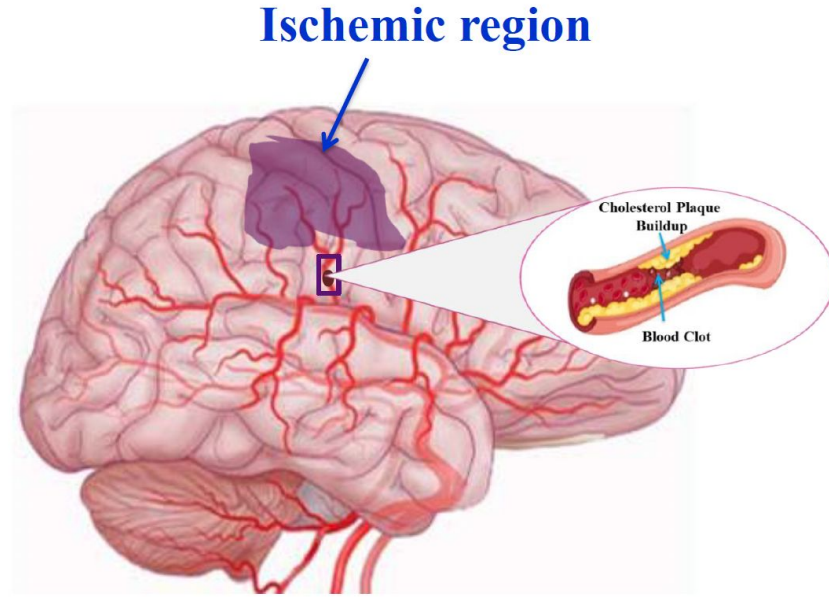


# Brain stroke

- An **ischemic brain** refers to a brain that has suffered from **ischemia**, which is a **condition where there is a lack of blood flow to a part of the brain**, causing oxygen and nutrient deprivation.

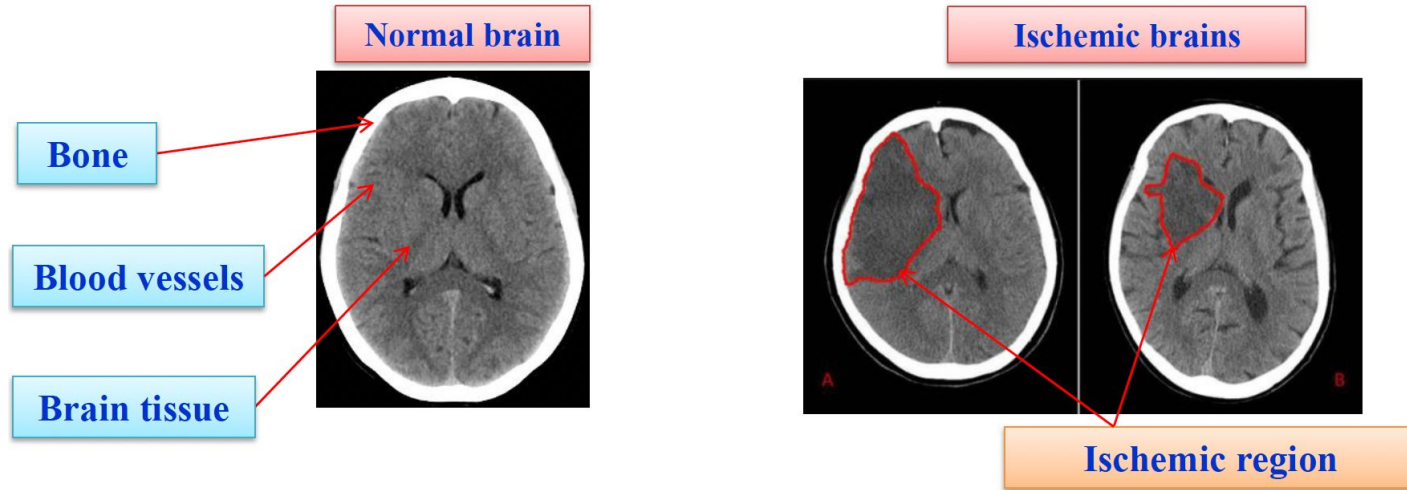


Normal Human Brain



Brain of Ischemic Stroke Patient

# Image of normal vs. ischemic brain

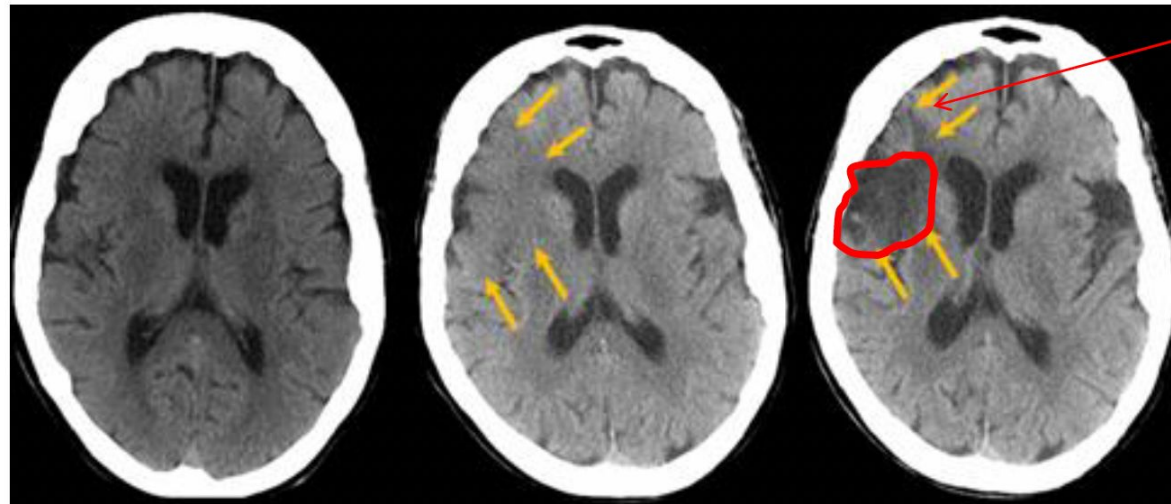


| Color difference between the parts of normal vs. ischemic brain |              |                             |
|-----------------------------------------------------------------|--------------|-----------------------------|
| Parts                                                           | Colors       |                             |
|                                                                 | Normal brain | Ischemic brain              |
| Bone                                                            | White        | White                       |
| Blood vessel                                                    | Mostly white | Mostly white                |
| Brain tissue                                                    | Mostly gray  | Dark gray (Ischemic region) |

# Normal progression of infarction

- Infarction

- Permanent tissue death from prolonged ischemia (irreversible)
- Difficult to find out in brain CT taken within hours of stroke!



3 hours

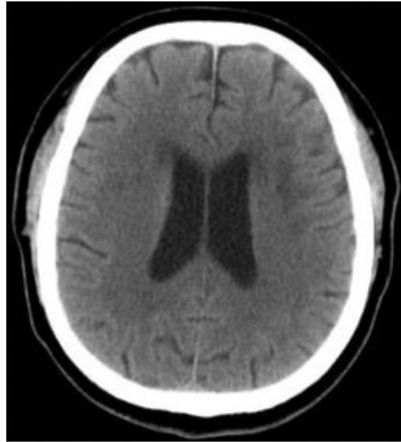
1 day

5 days

Ischemic region  
(the brain tissue  
color is changed  
to dark grey)

# Clinical difficulties

- **Difficult to distinguish** between normal and ischemic stroke patients with naked eye
  - **Similar color intensity** of brain tissue



**NCCT image: Normal patient**



**NCCT image: Ischemic stroke patient**

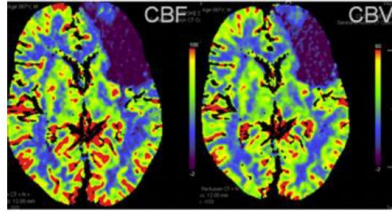
# Clinical difficulties

- **CTA or CTP**

- injecting a **contrast agent (dye)** into the bloodstream to visualize blood vessels in the brain and neck.
- Uses high rates of iodine-rich contrast material
  - Carcinogenic
  - Potential of allergic reactions
  - Health issues corresponding to radiation does



CTA

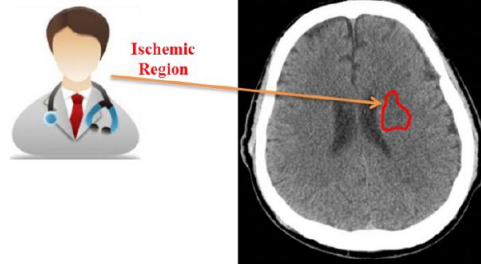
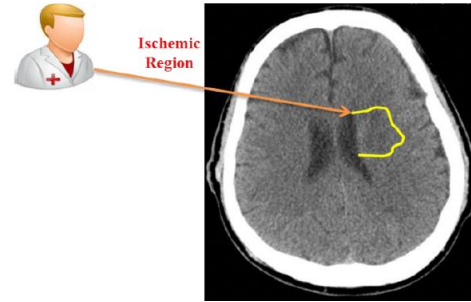
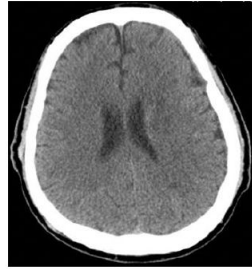


CTP



# Technical difficulties

- Localization of ischemic region is difficult!

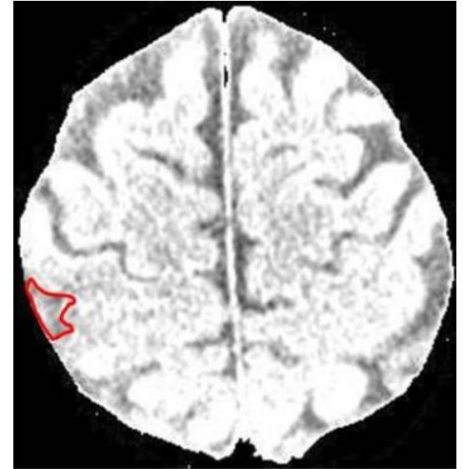
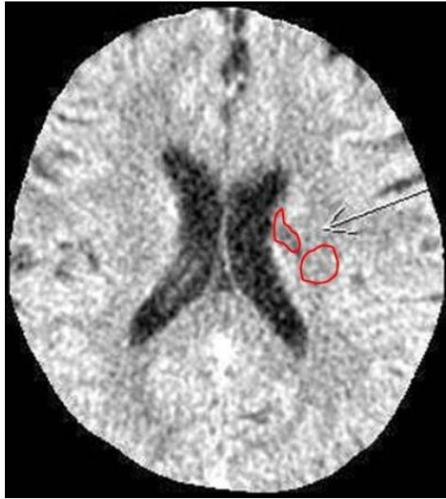


NCCT image: Ischemic stroke patient



# Technical difficulties

- Ischemic stroke has **no particular location!**



NCCT images: Ischemic stroke patients



# Ischemic stroke diagnosis using AI

- Classification of **normal** and **ischemic** stroke patients with deep learning
- **Localization** of ischemic stroke region with deep learning
- **Models to use**
  - AlexNet, VGGNet, GoogleNet, Inception, ResNet, etc.
  - Transfer learning
    - Use the models pretrained on a large corpus of nature images
    - Finetune it on the existing labeled data

# Training data

- The input data are **NCCT images**
- Ground truth are determined using **DWI**

